

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

C02: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 6

Total Marks:30

Annexure A

Scenario-Based

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Scenario-Based

1. Use Case Diagram Scenario

A **Smart Pharmacy Management System** enables customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage medicine inventory and user accounts. The interactions among system users are not clearly defined.

Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.

(5 Marks)

ANSWER

Actors

1. **Customer**
2. **Pharmacist**
3. **Administrator**
4. **Payment Gateway** (External System)

Use Cases

Customer

- Register/Login
- Search Medicines
- Upload Prescription
- Place Medicine Order
- Make Online Payment
- Track Order

Pharmacist

- Verify Prescription
- Dispense Medicines
- Update Order Status

Administrator

- Manage Medicine Inventory
- Manage User Accounts

- View Reports (Optional)

Payment Gateway

- Process Payment

Relationships

<<include>>

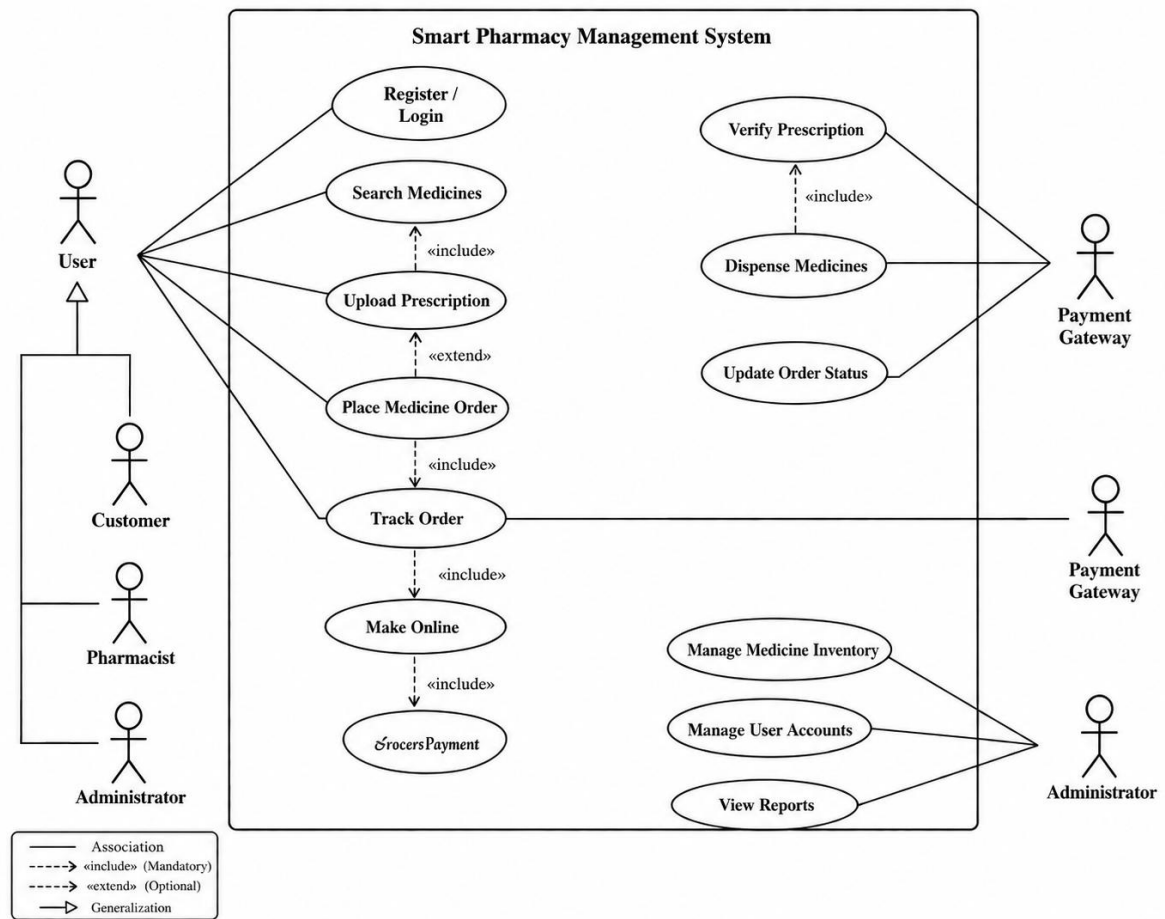
- **Place Medicine Order** <<include>> **Search Medicines**
- **Place Medicine Order** <<include>> **Make Online Payment**
- **Make Online Payment** <<include>> **Process Payment**
- **Dispense Medicines** <<include>> **Verify Prescription**

<<extend>>

- **Upload Prescription** <<extend>> **Place Medicine Order**
(*Only required when prescription medicines are ordered.*)

Generalization

- **User** (General Actor)
 - Customer
 - Pharmacist
 - Administrator



2. Class Diagram Scenario

A **Smart Hotel Management System** consists of entities such as Guest, Room, Reservation, Payment, and Receptionist. The existing design does not clearly define class attributes, operations, or relationships among these entities.

Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.

(5 Marks)

ANSWER

Smart Hotel Management System – Class Diagram

Classes with Attributes and Methods

1. Guest

Attributes

- guestId : int
- name : String
- phone : String
- email : String
- address : String

Methods

- register()
- login()
- bookRoom()
- cancelReservation()
- makePayment()

2. Room

Attributes

- roomId : int
- roomNumber : String
- roomType : String
- price : double
- status : String

Methods

- checkAvailability()
- updateStatus()
- assignRoom()

3. Reservation

Attributes

- reservationId : int
- checkInDate : Date
- checkOutDate : Date
- bookingStatus : String

Methods

- createReservation()
- modifyReservation()
- cancelReservation()

4. Payment

Attributes

- paymentId : int
- amount : double
- paymentMethod : String
- paymentStatus : String

Methods

- processPayment()
- generateReceipt()

5. Receptionist

Attributes

- employeeId : int
- name : String
- contact : String
- shift : String

Methods

- checkInGuest()
- checkOutGuest()
- assignRoom()
- manageReservation()

6. Person (Superclass)

Attributes

- id : int
- name : String
- phone : String

Methods

- login()
- logout()

Relationships

Inheritance (Generalization)

- **Person**
 - Guest
 - Receptionist

Association

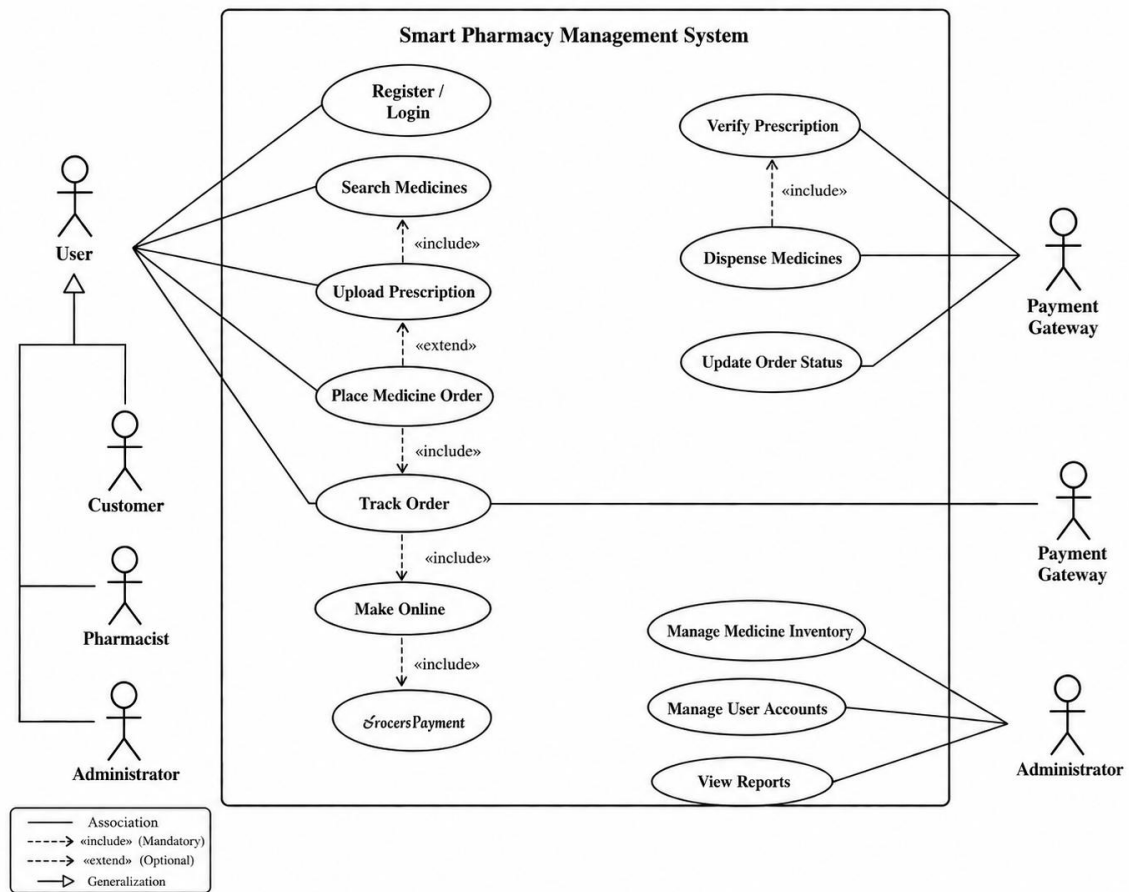
- Guest **books** Reservation
- Receptionist **manages** Reservation

Aggregation

- Reservation ◇— Room
 - A reservation contains one or more rooms, but rooms can exist independently.

Composition

- Reservation ◆— Payment
 - Payment exists only for a reservation.



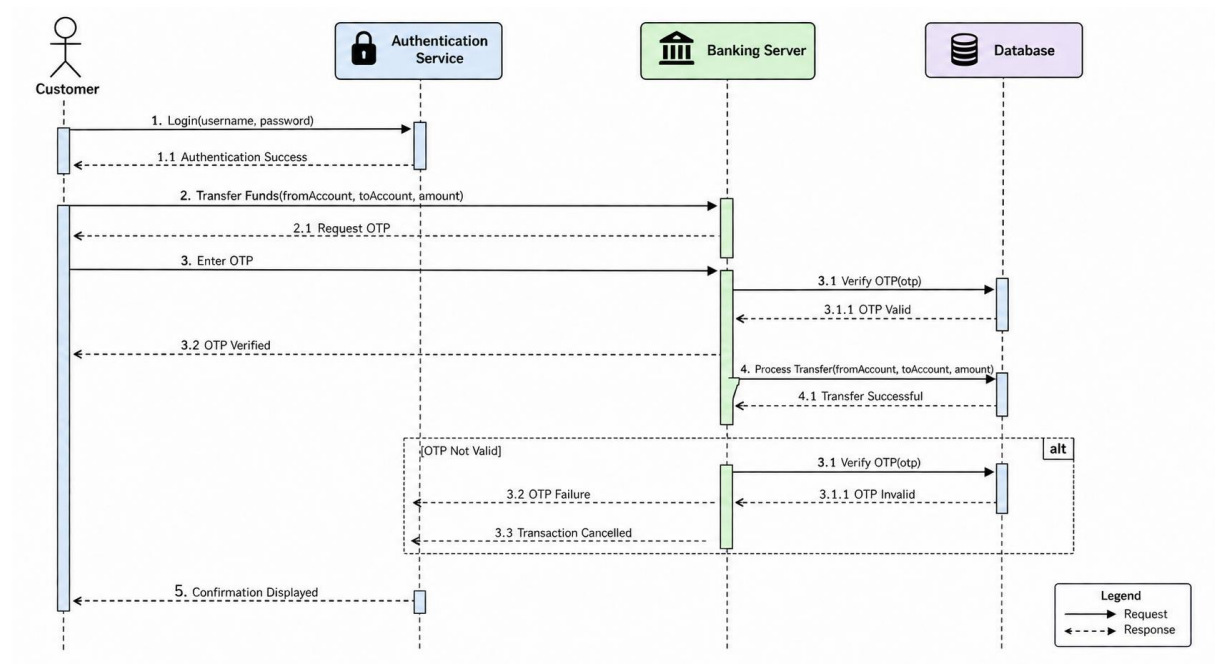
3. Sequence Diagram Scenario

In an **Online Banking System**, a customer logs in, transfers funds, verifies the transaction using OTP, and receives a transaction confirmation. The communication among the customer, authentication service, banking server, and database is incomplete.

Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.

(5 Marks)

ANSWER



4. Activity Diagram Scenario

A **Passport Application System** allows applicants to submit applications, upload supporting documents, complete identity verification, pay processing fees, and receive passports. The workflow does not clearly represent approval, rejection, or concurrent activities.

Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.

(5 Marks)

ANSWER

The **Activity Diagram** models the workflow of the **Passport Application System** from the beginning of the application process until the passport is issued or the application is rejected.

Step-by-Step Workflow

1. **Start**
 1. The passport application process begins.
2. **Applicant Registers / Logs In**
 1. The applicant creates an account or logs into the system.
3. **Submit Passport Application**
 1. The applicant fills in the passport application form and submits it.
4. **Upload Supporting Documents**
 1. The applicant uploads the required documents such as:
 1. Aadhaar Card
 2. Birth Certificate
 3. Address Proof
 4. Passport-size Photo
5. **Identity Verification**
 1. The system verifies the applicant's identity using the submitted documents.
6. **Decision Node: Verification Successful?**
 1. **Yes:** Continue to the next stage.
 2. **No:** The system requests additional information or corrected documents, and the applicant uploads them again.

Fork Node (Parallel Activities)

After successful verification, the workflow splits into **two parallel activities**.

Activity 1

- Upload any remaining supporting documents (if required).

Activity 2

- Pay Passport Processing Fees.
- The system processes the payment.

Both activities happen **simultaneously**.

Join Node

- The workflow waits until **both parallel activities are completed**.
- After both finish, they merge at the **Join Node**.

Application Review

- The passport office reviews:
 - Application form
 - Uploaded documents
 - Identity verification
 - Payment status

Decision Node: Application Approved?

If YES

1. Passport is issued.
2. Passport is dispatched.
3. Applicant receives a notification.
4. Process ends.

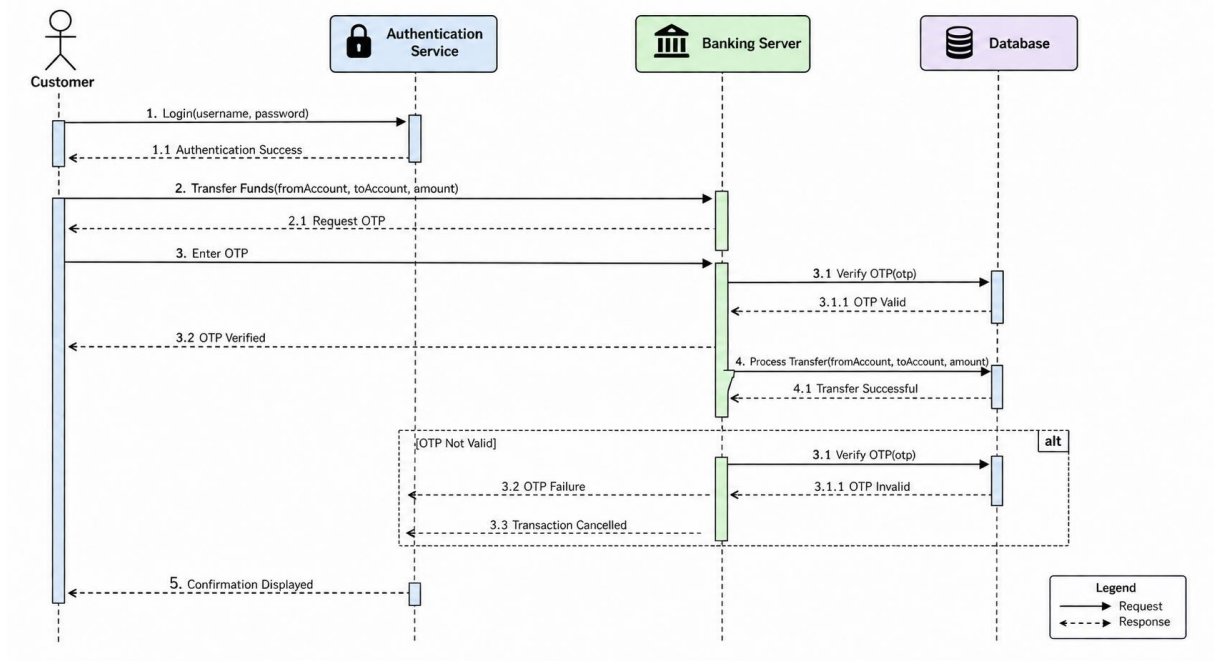
If NO

1. Application is rejected.
2. Applicant is notified of the rejection.
3. Process ends.

End

The activity diagram ends after either:

- Passport is successfully issued and dispatched, **or**
- The application is rejected.



5. State Diagram Scenario

A **Smart Elevator Control System** operates through states such as Idle, Door Open, Door Closed, Moving Up, Moving Down, Emergency Stop, and Maintenance Mode. The transitions between these states are not clearly specified.

Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the elevator's behavior.

(5 Marks)

ANSWER

A **State Diagram** shows the different **states** of an object (elevator) and the **events** that cause it to move from one state to another. It represents the complete behavior of the Smart Elevator Control System.

States in the Diagram

1. **Idle**
 1. The elevator is waiting for a request.
 2. This is the default state after the system starts or completes a trip.
2. **Door Open**
 1. The elevator doors are open.
 2. Passengers can enter or exit.
3. **Door Closed**
 1. The doors are closed.
 2. The elevator is ready to move if a floor is selected.
4. **Moving Up**
 1. The elevator moves upward to the selected floor.
5. **Moving Down**
 1. The elevator moves downward to the selected floor.
6. **Emergency Stop**
 1. Triggered when the emergency button is pressed.
 2. The elevator immediately stops and waits until the emergency is reset.
7. **Maintenance Mode**
 1. Used for servicing or repairing the elevator.
 2. Normal operation is disabled until maintenance is completed.

Step-by-Step Working

1. Initial State → Idle

- The system starts.
- The elevator enters the **Idle** state and waits for user input.

Transition:

Start System → Idle

2. Idle → Door Open

- A user calls the elevator or presses the Open Door button.
- The doors open for passengers.

Trigger:

- Call Elevator
- Open Button

3. Door Open → Door Closed

- After passengers enter or leave, the timer expires or the Close button is pressed.
- The doors close.

Trigger:

- Close Button
- Timer Expired

4. Door Closed → Moving Up

- A floor above the current floor is selected.
- The elevator starts moving upward.

Trigger:

- Floor Above Selected

5. Door Closed → Moving Down

- A floor below the current floor is selected.
- The elevator starts moving downward.

Trigger:

- Floor Below Selected

6. Moving Up → Door Open

- The elevator reaches the selected floor.
- The doors open for passengers.

Trigger:

- Destination Reached

7. Moving Down → Door Open

- The elevator reaches the requested lower floor.
- The doors open.

Trigger:

- Destination Reached

8. Door Closed → Idle

- If there are no pending requests, the elevator returns to the **Idle** state.

Trigger:

- No Request

9. Emergency Stop

The emergency button can be pressed while the elevator is in:

- Door Open
- Door Closed
- Moving Up
- Moving Down

All these states transition to **Emergency Stop**.

Trigger:

- Emergency Button Pressed

10. Emergency Stop → Idle

- After the emergency is resolved and reset, the elevator returns to **Idle**.

Trigger:

- Reset Emergency

11. Idle → Maintenance Mode

- The administrator or technician enables maintenance mode.
- The elevator stops serving passengers.

Trigger:

- Maintenance Enabled

12. Maintenance Mode → Idle

- After servicing is completed, the elevator returns to normal operation.

Trigger:

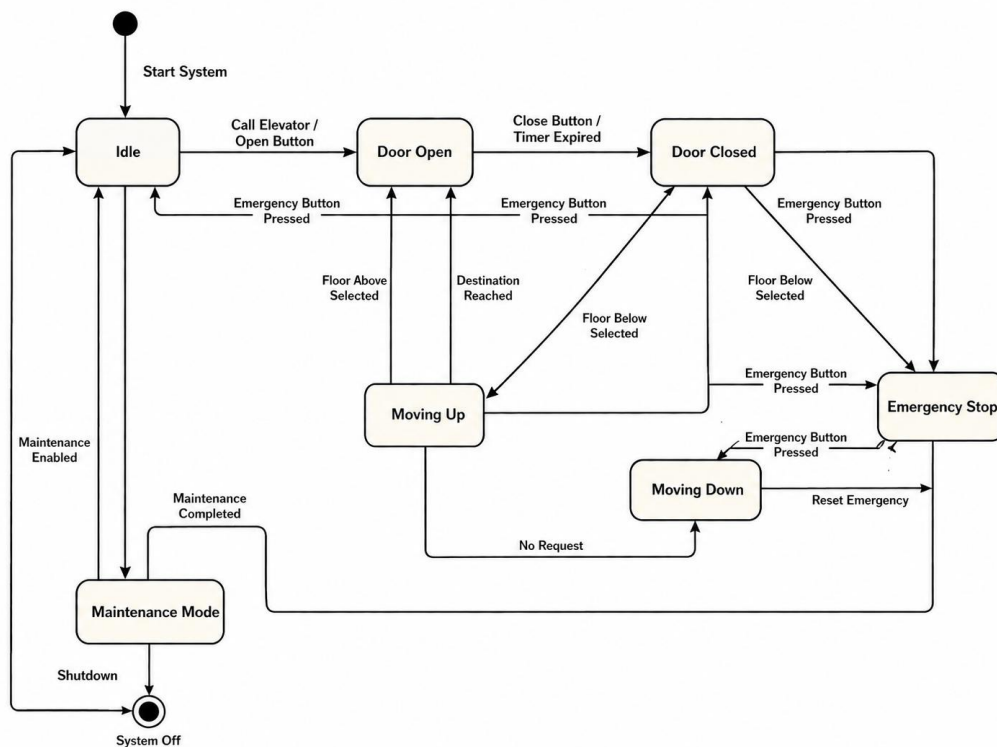
- Maintenance Completed

13. Idle → Final State

- When the system is shut down, the elevator reaches the **Final State**.

Trigger:

- Shutdown



6. Deployment Diagram Scenario

A **Smart Home Automation System** uses smart sensors, IoT controllers, a home gateway, cloud servers, and a mobile application to monitor and control lighting, security, and appliances remotely. The deployment architecture does not clearly illustrate hardware nodes, deployed software, or communication channels.

Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.

(5 Marks)

ANSWER

A **Deployment Diagram** represents the **physical architecture** of the Smart Home Automation System. It shows the **hardware devices (nodes)**, the **software deployed on each device (artifacts)**, and the **communication links** between them.

Hardware Nodes

1. Smart Sensors Node

This node contains all the smart sensors installed in the house.

Examples

- Temperature Sensor
- Motion Sensor
- Smoke Sensor
- Door Sensor
- Light Sensor

Software Artifact

Sensor Firmware

- Collects sensor data.
- Detects environmental changes.
- Sends data to the IoT Controller/Home Gateway.

2. IoT Controller Node

This is the controller that manages all smart devices.

Software Artifact

IoT Controller Software

Functions:

- Controls home appliances.
- Executes automation rules.
- Manages connected devices.

Example:

- If motion is detected, switch on the light.
- If smoke is detected, activate the alarm.

3. Smart Appliance Node

Represents the smart devices inside the home.

Examples:

- Smart Lights
- Air Conditioner
- Fan
- Smart Door Lock

Software Artifact

Appliance Controller

Functions:

- Receives commands from the IoT Controller.
- Turns appliances ON/OFF.
- Reports appliance status.

4. Home Gateway

The Home Gateway connects the local smart home network to the Internet.

Software Artifact

Gateway Software

Functions:

- Collects data from sensors.
- Communicates with cloud servers.
- Provides device security.
- Manages local communication.

5. Cloud Server

The cloud server stores and processes all smart home data.

Software Artifact

Cloud Services

Functions:

- User Authentication
- Data Storage
- Automation Processing
- Notification Services
- Remote Access

6. Mobile Application

The user controls the smart home using a mobile phone.

Software Artifact

Mobile App

Functions:

- User Login
- Monitor Devices
- Control Appliances
- Receive Notifications

Communication Links

Smart Sensors → Home Gateway

Communication Protocols:

- Wi-Fi
- ZigBee
- Bluetooth

Purpose:

- Transfer sensor readings to the Home Gateway.

IoT Controller → Home Gateway

Communication:

- Wireless communication (Wi-Fi/ZigBee/Bluetooth)

Purpose:

- Sends automation commands and receives sensor data.

Home Gateway → Cloud Server

Communication:

- **HTTPS/TLS**

Purpose:

- Securely transfers home data to the cloud.
- Protects user information using encryption.

Cloud Server → Mobile Application

Communication:

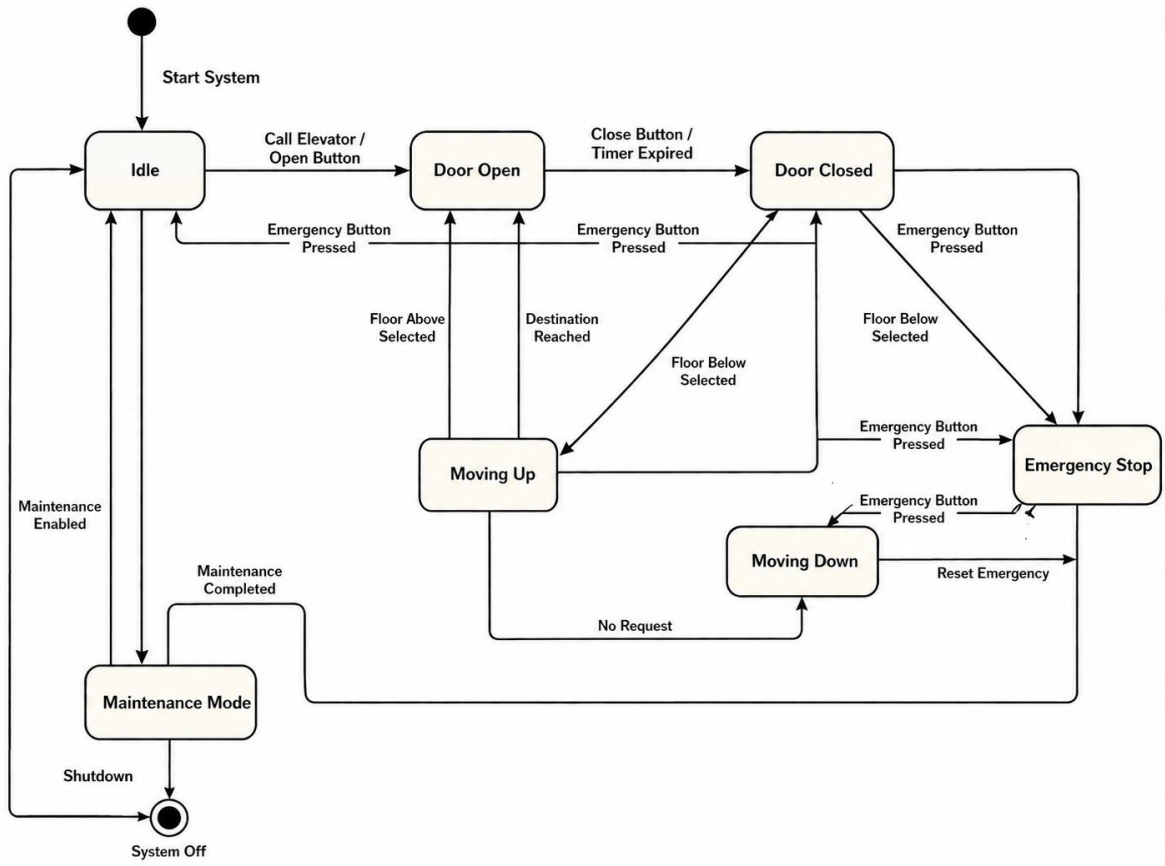
- **HTTPS/TLS**

Purpose:

- Sends notifications.
- Allows remote monitoring and control.

Overall Workflow

1. Sensors continuously monitor the home environment.
2. Sensor data is sent to the **Home Gateway** through wireless communication.
3. The **IoT Controller** processes the data and controls smart appliances.
4. The **Home Gateway** securely sends data to the **Cloud Server** using HTTPS/TLS.
5. The **Cloud Server** stores data, processes automation rules, and manages user accounts.
6. The **Mobile Application** communicates with the Cloud Server, allowing users to monitor and control lighting, security, and appliances remotely.



RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand